

Preliminary thesis proposal

Interpretable DAW-State Graphs for Prediction and Explanation in AI-Assisted Music Production

Preliminary thesis proposal for the PhD position at the Music Technology Group (Universitat Pompeu Fabra) with Steinberg Media Technologies GmbH as industrial partner.

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1. Motivation and problem statement

Digital audio workstations are the central instrument of contemporary music production, yet the knowledge they contain is largely invisible to current AI music systems. A DAW session encodes a producer's craft as *state*: track organization, routing topology, insert and send effect chains, parameter settings, and the arrangement of clips and events. Most assistive technologies for music production operate downstream of this state — on rendered audio — or upstream of it, on isolated plugin parameters. The session itself, where production decisions actually live, remains an unmodeled black box.

This thesis addresses the topic of the call directly: **representations of the states of a DAW, such as audio effects or mixing details, and how they influence the final acoustic outcome**, with the goal of providing and evaluating **methods that allow prediction of such DAW states** to support the creativity of musicians and producers.

The central claim is that DAW state should be modeled as a **typed, partially observable graph** — a representation in which tracks, effect chains, sends, returns, parameters, and their relations are explicit, nameable objects. Such a representation supports three things that end-to-end audio models do not: (i) *prediction at the level producers act on* (which device, which routing, which setting — not which waveform), (ii) *explanation by construction* (a predicted state change can cite the session structure that motivated it), and (iii) *honest handling of partial observability* (proprietary formats and plugin-internal state bound what any system can see, and the representation should say so).

2. Research questions

RQ1 — Representation. What graph-structured representations of DAW session state are expressive enough to capture production-relevant structure (chains, routing, arrangement) while remaining DAW-agnostic — instantiable from Cubase/VST3 session structures as well as other DAW paradigms — and honest about unobserved state?

RQ2 — Prediction. Given a partial session state and optionally audio descriptors of its material, how well can models predict withheld or future DAW states: the composition of a device chain, likely parameter regions, the next production action in an edit sequence? Which model classes (interpretable probabilistic baselines, graph neural networks, sequence models over edit histories) offer the best accuracy/interpretability trade-off?

RQ3 — Outcome linkage. How do DAW-state changes map to measurable acoustic outcomes? Can models predict descriptor-space consequences of state edits (e.g., the spectral and dynamic effect of an insert chain), and conversely, infer plausible states from audio evidence?

RQ4 — Human-centered explanation. When state predictions are surfaced to producers as suggestions, what explanation forms (graph citations, precedent examples, counterfactuals) support appropriate trust, and how do such systems affect producer agency and creative outcomes, measured through user studies?

3. Methodology and work plan

Year 1 — Representation and data. Formalize the typed session-state graph schema (building on the preliminary prototype below), define the Cubase/VST3 instantiation with Steinberg — tracks, events, insert/send effects, and the VST3 parameter model give the schema a concrete industrial grounding — and construct datasets. Three complementary sources: (a) synthetic corpora generated from parameterized production templates, useful for controlled prediction benchmarks; (b) open multitrack resources (MedleyDB, the Cambridge-MT multitrack library, MUSDB18) paired with reconstructed or annotated session structures; (c) in collaboration with Steinberg, anonymized real session structures and edit histories where feasible — the industrial partnership’s distinctive opportunity. Audio material is descriptor-encoded with Essentia; Freesound provides loop/sample-level material with community metadata.

Year 2 — Predictive models. Develop and compare models for the RQ2 tasks in increasing order of complexity: interpretable probabilistic baselines (co-occurrence and conditional models over chain composition), graph neural networks over the typed session graph (masked-node and link prediction), and sequence models over edit histories (next-action prediction). Evaluate against the RQ3 linkage: joint models of state and descriptor evidence. Every learned model is paired with an explanation mechanism, and faithfulness of explanations is evaluated alongside accuracy.

Year 3 — Human-centered evaluation and synthesis. Integrate the best models into an assistive prototype (suggestion surfaces inside or alongside a DAW workflow, with Steinberg’s product context) and run producer studies: explanation quality, trust calibration, perceived agency, and effect on production outcomes. Synthesis, open releases, and thesis writing.

Throughout: open-science practice — public code, documented schemas, released datasets where licensing allows — consistent with MTG norms.

4. Preliminary work

I have built and released a working prototype, **Session State Explorer v0** (github.com/colonelkernel/AbletonSessionStateExplorer), that implements the representation-and-explanation half of this proposal at small scale:

- a typed, DAW-agnostic session-state model (tracks, clips, scenes, devices, parameters, sends, returns, master) instantiated in **two DAW dialects** — an Ableton-style demo (session grid) and a Cubase-style demo (linear arranger, wired FX-channel sends, dialect-supplied device families) — as direct evidence for the cross-DAW representation claim of RQ1;
- compilation of session state into a typed directed graph (11 node types, 12 edge types) with graph-level statistics and explicit modeling of placeholder/uncertain elements;
- audio descriptor extraction (librosa; Essentia and pyloudnorm as optional backends) attached to graph entities;
- six rule-based recommendations that are explainable by construction — each cites the graph nodes it reasons over, states a confidence, and carries an explicit caveat preserving producer agency;
- a first learned proof-of-concept for RQ2: a masked device-family prediction baseline trained on a seeded synthetic session corpus, with honest reporting of its synthetic-data scope;
- a structural **session diff** between versions (devices, sends, returns, tempo, parameters) whose built-in demonstration enacts the system's own recommendations and verifies them as state changes — a working seed of RQ3's version-comparison methodology;
- a **Live extension** built on Ableton's public Extensions SDK that exports real session state from inside the DAW into the model's schema — the observation pathway the thesis proposes, working end-to-end on the Ableton side and mirroring what first-party Cubase instrumentation would provide with Steinberg;
- cautious surface inspectors for both Ableton `.als` files and Cubase Track Archive XML, and a documented export adapter — partial observability as a designed-for property on both DAW sides, rather than an afterthought.

The prototype is deliberately modest — its purpose is to make the research object concrete and to establish the explanation contract that learned models in the thesis must satisfy.

5. Fit with the supervisory team

The proposal is designed at the seam the supervisory team spans. Essentia (Bogdanov) is the descriptor backbone for RQ3's state–outcome linkage; Freesound and community audio data (Font) ground the loop/sample layer of session material; the open, transparent AI-for-music agenda (Serra) is the methodological frame for RQ4. On the Steinberg side, MIR-informed production tooling (Wolff) and the VST3/SDK and product perspective (Rolland) make the Cubase instantiation of RQ1 and the Year-3 integration realistic rather than speculative.

6. Responsible and human-centered AI

Assistive production systems risk homogenizing style and eroding producer agency. The design position of this thesis is that predictions are surfaced as *inspectable options with stated evidence and stated limits*, never as corrections; that models disclose the observability boundary they operate within; and that evaluation includes agency and trust, not only accuracy. This aligns with the MTG’s stated commitment to trustworthy and transparent AI for music.

7. Selected references

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This is a preliminary proposal; the work plan is expected to be refined with the supervisory team during the first months of the PhD.